Euler's Equations: Symbolic versus Numerical Mathematics

Johan Hoffman Johan Jansson Claes Johnson

KTH Royal Institute of Technology, Stockholm

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Abstract

Three things are to be kept apart: a fluid flow, which is a phenomenon; Euler's equations, which are a model of it; and a proof about the equations, which is a statement about the model. We state the equations, note that they are parameter free, and observe that the mathematical question asked of them — existence, regularity, uniqueness — has not been answered by analysis in symbolic form, because the solutions turn out to have a complicated nature which symbolic analysis has no grip on. Solutions in digital form over a mesh can be computed by a model we name *Numerical Euler* — a stabilised finite element discretisation together with duality-based output error control and adaptive refinement — and they are turbulent, fluctuating over a range of scales, with well determined mean values such as drag and lift which agree with measurement and arrive with a computed bound on their error. We then ask what a recent symbolic proof of finite-time blowup establishes, and conclude that it is a statement about the model which does not reach either of the other two levels: it exhibits no individual case, so nothing can be inserted into the equations, inspected, or measured.

1 Euler, 1755

In 1755 Leonhard Euler presented to the Berlin Academy the memoir *Principes généraux du mouvement des fluides* [4], in which he wrote down the equations of motion of an incompressible fluid. He took them to contain the whole of the mechanics of fluids, and he was explicit that what then stood in the way of using them was not physics but mathematics:

> And if it is not permitted to us to penetrate to a complete knowledge concerning the motion of fluids, it is not to mechanics, or to the insufficiency of the known principles of motion, that we must attribute the cause. It is analysis itself which abandons us here.

What Euler had in mind was *ideal* fluid flow — inviscid incompressible flow — as a model of the slightly viscous, nearly incompressible flow of real fluids such as water and air at subsonic speeds. He formulated his model in the following form, named *Euler's equations*:

$$\dot{u} + (u \cdot \nabla)u + \nabla p = f \qquad \text{in } \Omega, \tag{1}$$

$$\nabla \cdot u = 0 \qquad \text{in } \Omega, \tag{2}$$

$$u \cdot n = 0 \qquad \text{on } \Gamma. \tag{3}$$

Here $\Omega \subset \mathbb{R}^3$ is the domain occupied by the fluid and Γ its boundary, $x = (x_1, x_2, x_3)$ is a Euclidean coordinate and t time; $u(x,t) \in \mathbb{R}^3$ is the velocity and $p(x,t)$ the pressure of a fluid of unit density; f is a given force, n the outward normal to Γ, and $u(\cdot, 0) = u^0$ a given initial velocity. The equations hold for $0 < t \leq T$.

What they say is the following. Equation (1) is Newton's second law: for unit density the mass of a fluid particle is its volume, so force equals acceleration directly, the left-hand side being the acceleration of a particle following the flow and the right-hand side the force on it — the pressure gradient, which is what the surrounding fluid exerts, together with any body force. Equation (2) states incompressibility, the volume of a particle not changing, with the pressure as whatever field enforces it. Condition (3) states that the fluid does not pass through the wall, and nothing more: the normal component is fixed by impermeability, nothing in the equations determines the tangential velocity there, and prescribing it as well would over-determine the problem. There is no viscosity anywhere, and therefore no boundary layer.

The model is parameter free. There is no constant in (1)–(3) to be measured, fitted or chosen. What enters as data is the *geometry* — the domain, which is to say the shape of the body the fluid flows past — the *forcing*, and the initial state. Nothing else. Whatever the equations predict is a consequence of the geometry and the forcing alone, and any disagreement with observation is a fault of the model and cannot be repaired by adjustment.

2 d'Alembert's paradox, and the split

The expectation that the mechanics of fluids was contained in two short mathematical equations met an objection which was in print before the equations were. Three years earlier, in the *Essai d'une nouvelle théorie de la résistance des fluides* of 1752 [8], d'Alembert had shown that a body moving steadily through an ideal fluid experiences *no resistance at all*. Ships meet resistance and cannonballs are slowed, so the conclusion is absurd — and it is a conclusion about precisely the solutions of (1)–(3), to which it duly attached itself.

It deepened rather than dissolved. In the nineteenth century complex analysis supplied a machinery generating ideal flow as *potential flow* for any body one cared to draw, all of it smooth, all of it elegant, and all of it with zero drag. The better the mathematics got, the larger grew the family of solutions describing nothing. Something was wrong with the equations and nobody could say what.

What it cost. The consequence was not a puzzle set aside for later. It was a divorce. Mathematical fluid mechanics, unable to predict the resistance of a body moving through water, became a subject whose central prediction was known to be false — which is to say a joke, and it was treated as one by the people who had to build ships and, later, aircraft. The practical science went its own way as *hydraulics*: correlations, loss coefficients, model tests in towing tanks and wind tunnels, and a working knowledge of turbulent flow that owed nothing to Euler's equations and could not be derived from them. The division is well known and was put best as a distinction between engineers who observed what could not be explained and mathematicians who explained what could not be observed.[1]

[1] The remark is usually credited to Hinshelwood and is quoted by Lighthill. [TO VERIFY.]

So the two halves of the subject parted. Real fluids were understood empirically and without mathematics; the mathematics was exact, beautiful and about nothing.

3 Leray, 1934: turbulent solutions without uniqueness

Nothing in the mathematics moved this position for the better part of two centuries. What moved it was Leray, in the paper of 1934 on the motion of a viscous liquid filling space [5]. Leray abandoned the requirement that a solution be smooth and defined a *weak* solution, satisfying the equations in an integrated sense against test functions. For such solutions he proved global existence. And he named them for what he took them to represent: *solutions turbulentes*.

This was the decisive step and it did not finish the job. What Leray obtained was existence without *uniqueness*, and uniqueness is what a model needs if it is to predict anything. A theory which says that solutions exist, and cannot say which one a given body in a given stream will produce, determines no drag and no lift. The subject gained a solution concept adequate to turbulence and still could not deliver a number, and the uniqueness question has remained open since — for weak solutions of Euler it is now known to be *false*.

4 Numerical Euler: weak solutions with residual control

The gap Leray left is where the present account begins. It was closed, in our view, not by proving uniqueness of the solution — which is unavailable, and false — but by changing what uniqueness is asked of. The solution concept set out in [7] is a weak solution together with *control of the residual*, and what it delivers is *uniqueness of mean-value outputs*: the flow field is not determined, and the drag and the lift are.

We name the computational model **Numerical Euler**. It has three parts and the third is not an afterthought:

1. the equations (1)–(3), discretised by a stabilised finite element method over a mesh [7], the stabilisation acting as a vanishing artificial viscosity, so that what is computed is a viscosity solution of Euler, the limit taken as the mesh is refined;

2. *duality-based output error control*: for each output of interest — a drag, a lift — the error is bounded by the residual of the computed solution weighted by the solution of an adjoint problem, which measures how much the residual matters for that output;

3. *adaptivity*: the mesh is refined where the weighted residual says the error lives, and not uniformly, until the bound on the output is below what is asked.

The error control is part of the model rather than an assessment of it. What Numerical Euler returns is not a flow but a *pair*: an output and a bound on how wrong that output may be, computed together. A model returning the first without the second would be a weaker thing, since nothing could be concluded from it.

And it is a solution of Euler. That the limit satisfies (1)–(3) weakly is not an identification offered because the numbers look right. The stabilisation is a weighted least-squares control of the residual; the discrete energy estimate bounds that residual in a weighted norm; testing against a smooth function sends it to zero in the sense the weak formulation requires. The limit is a weak solution in Leray's sense, as a theorem — and unlike Leray's, it comes with a statement of what is determined.

What it produces, none of it put in. The computed flow is *turbulent*: fluctuating in time and over a range of scales which widens as the mesh is refined, with no symmetry and no steady state. Its *mean values are determined*: drag, lift, and the position of the separation line converge under refinement, each arriving with its bound. And its *velocity stays bounded while its gradients do not*, the solution being Hölder continuous with exponent about $1/3$ — the exponent below which Onsager observed that energy conservation can fail [15] — and dissipating energy at a rate independent of the vanishing viscosity.

The viscosity introduced, and why its size does not matter. A computation at finite resolution cannot have zero dissipation, so Numerical Euler necessarily introduces a viscosity. This is the obvious objection to it, and the answer is a computational fact rather than an argument: *the mean-value outputs do not depend on the absolute size of the viscosity introduced.* Refine the mesh and the artificial viscosity falls, the resolved scales multiply, the flow field changes in every detail — and the drag does not move. The dissipation rate does not fall with the viscosity either; it settles at a value fixed by the geometry and the free stream. That independence is the dissipation anomaly in computational form, and it places the computation and the fluid in the same position rather than in different ones: air has a small viscosity whose absolute size does not determine the drag of a wing, and Numerical Euler has a small viscosity whose absolute size does not determine the computed drag. Neither is inviscid and neither needs to be.

d'Alembert's paradox, resolved. With turbulent solutions available for inspection the paradox can be answered, and was [9]. Potential flow is a solution and it is *unstable* — stable where the flow attaches on the forward surface and accelerates, unstable where it separates on the rear surface and is retarded. The instability generates streamwise vorticity which stretches and breaks down into a turbulent wake, and the wake carries the drag. The same mechanism applied to a wing gives lift with no circulation condition imposed by hand [10]. The equations were never wrong. What was wrong was the assumption that the solution of interest is a smooth one, and two centuries of symbolic mathematics had been spent elaborating exactly the solutions that do not occur.

5 FEniCS, automation, and the workshops

A solution concept is of no use to an engineer who cannot compute with it, and the step from the concept to a tool was automation. The FEniCS project [6] automates the step from a variational form to compiled code: the equations are stated, the discretisation is generated, and the adjoint problem required by the error control is generated with them. What had been

a research computation became a method that could be run by somebody who had not written it.

The result has been taken into use, and the place to see whether it works is not our own publications but the benchmark workshops, where the cases, the geometries and the measurements belong to the organisers and predictions are submitted before the measurements are released. Numerical Euler has been put through them: the high-order CFD workshop case CS1 [12], and validation against the NASA High Lift Prediction Workshop 4 [13]. Drag within a few per cent, the separation line where experiment puts it, the lift curve and the stall, the drag crisis [11].

That is the end of the arc begun in 1752. The subject which split into a mathematics that explained what could not be observed and a hydraulics that observed what could not be explained has a model with no parameters, solved from the equations, delivering the numbers an aircraft designer asks for, to a tolerance the computation itself states, against measurements held by somebody else.

6 Inspection of blow-up proofs by OpenAI

Against this stands a recent proof, produced by an AI system and formalised in Lean, that there exists smooth, compactly supported, divergence-free initial data on $\mathbb{R}^3$ whose smooth Euler solution has finite maximal lifespan: the velocity gradient becomes unbounded, and the Beale–Kato–Majda [14] vorticity integral diverges [1]. Comparable results were being prepared independently and by hand [2]. We take the proof to be correct, and nothing below depends on whether it is.

It is a statement about the *model*, and it is a real one: a question open for decades is settled. Three observations place it.

What blows up is the gradient, and that is what turbulence does too. The word blowup invites the reading that the velocity becomes infinite. It does not, and the theorem does not claim it. What is asserted is

$$\limsup_{t\uparrow T_*} \|\nabla u(t)\|_{L^\infty} = \infty, \qquad \int_0^{T_*} \|\operatorname{curl} u(t)\|_{L^\infty}\, dt = \infty,$$

the gradient and the vorticity integral, with nothing said about the velocity becoming unbounded.

That is precisely the signature of the computed turbulent solution of Section 4: gradients diverging while the velocity stays bounded. The two accounts therefore agree on the fact, that smooth solutions of (1)–(3) break down in finite time with unbounded gradients, which is what we reported computationally in [3]. The agreement is larger than the vocabulary of blowup against turbulence suggests, and it is worth stating before the differences.

But the same quantity diverges for opposite reasons. In the turbulent flow the gradient grows because the cascade carries energy to smaller and smaller scales and *the energy leaves*; the divergence of ∇u is the mechanism of dissipation, which is the drag. In the construction the gradient grows because the energy *cannot* leave — the solution is classical and

conserves it exactly — so a growing gradient can be accommodated only by concentrating what is there. One divergence is energy departing, the other is energy piling up. The symptom is shared and the physiology is opposite.

They differ also in kind of event. In the turbulent solution the unboundedness of the gradient is a *persistent condition*: the solution goes on existing, as a weak solution, dissipating, for as long as one cares to run it. In the construction it is a *terminal event*: at T_* the smooth solution ceases, and the theorem says nothing about what follows.

Which locates the result rather exactly. T_* is the end of the laminar regime. That is a real thing to have established, and we would not have it dismissed: the breakdown of smooth Euler flow in finite time was asserted on computational evidence and is now proved, for one constructed datum. But a fluid spends essentially none of its life before T_* and essentially all of it after. The flows of Section 4 — a wake, a separation line, a drag — are what happens beyond the breakdown, and they are what the equations were written to describe. The theorem stops at the point where fluid mechanics begins.

It conserves energy, and therefore cannot be a fluid. With smooth data the solution is classical up to the blowup time, and a classical solution of (1)–(2) with decaying data satisfies

$$\frac{d}{dt}\int \tfrac{1}{2}|u|^2 = 0,$$

by an integration by parts. The construction therefore dissipates *nothing* over the whole of its existence. The singularity is not a mechanism by which energy is lost but one by which it is piled up: at the singular time all of it is still present, gathered into a vanishing region. A fluid loses energy continuously from the moment it meets a body, and the loss is the drag.

It exhibits no individual case. The datum is the limit of an iteration over stages and has no closed form. Nothing can be plotted, nothing measured. Nor can either check on a *result* be applied: there is no symbolic expression to insert into (1)–(2) and reduce, and no numerical field either, since no finite stage of the iteration blows up — the property belongs to the limit. What exists is the existence.

What is proved was already claimed. There is a further observation, and it is not ungenerous. The theorem asserts that *some* smooth datum has finite lifespan. Our position, stated in [3] in 2008 and resting on computation, is stronger: *every* laminar Euler flow at high Reynolds number is unstable and breaks down, with the velocity staying bounded while the gradients do not. The existence statement follows from ours as a corollary; if all smooth solutions break down then certainly one does.

What is new is therefore the proof and not the fact. That is a genuine contribution — a fact asserted on computational evidence is not the same as a fact established — and we do not diminish it. But it should be seen for what it is: a demonstration, in a special constructed case, of a phenomenon which is generic, which has been exhibited computationally for two decades, and which the instability of Section 4 predicts of every flow past every body.

And the viscous case does not say what it is taken to say. Two things about the companion proof for Navier–Stokes, offered without disputing it.

The first is a matter of what was proved. Fefferman's four alternatives do not form two pairs of negations: (A) and (B) are posed with $f \equiv 0$ and quantify over *all* admissible data, while (C) and (D) ask for the existence of admissible data *together with an admissible smooth forcing* for which smoothness fails. A proof of (C) therefore leaves (A) untouched. It does not show that a fluid tears itself into a singularity; it shows that *something can be done to a fluid* which does. Those are different statements and only the first would be about fluid flow.

The second is a matter of scale. Navier–Stokes rescales: if v solves the equations at unit viscosity with forcing g, then

$$u(x,t) = a\,v(bx,ct), \qquad a = \nu b, \quad c = ab, \quad f = a^2 b\,g, \tag{4}$$

solves them at viscosity ν, and the map is onto — one unit-viscosity construction generates the whole family. Running it down to the viscosity of a fluid then forces a choice, and there are only two. Hold the solution at size one, $a = 1$: the forcing amplitude is $a^2 b = \nu^{-1}$, so the force required diverges as the viscosity falls and leaves any fixed admissible class. Or hold the forcing, $a^2 b = 1$: then

$$\text{amplitude } \nu^{1/3}, \qquad \text{length } \nu^{2/3}, \qquad \text{time } \nu^{1/3},$$

all tending to zero — at $\nu = 10^{-6}$ the velocity reaches 10^{-2} on a scale of 10^{-4} before blowing up.

Set that beside the flow the problem is about. Turbulence at small viscosity has amplitude *fixed* and structure refining. The constructed family has amplitude *vanishing* like $\nu^{1/3}$, which is the opposite behaviour, or else a forcing diverging like ν^{-1}, which is not a fluid at all. Neither branch survives the limit and there is no third. An independent exclusion of the mechanism under analytic forcing points the same way [17].

So the construction requires either a force no fluid is subject to, or a singularity too small to constitute anything, and Horace supplies the description: *parturient montes, nascetur ridiculus mus.*

Forwards and backwards. The two constructions of this paper are of the same formal kind and opposite in direction, and it is worth saying so plainly. The singular solution fixes the singularity wanted and solves *backwards* for data and a force that will produce it, the force being whatever the residual requires. Numerical Euler fixes the data a flow actually has and solves *forwards*, the residual becoming the bound on what may be asked of the result. Both are automated and both exhibit something. Only the second exhibits a fluid.

7 Three levels, and what reaches which

The title names three things and the point of the paper is that they are three.

A fluid flow is a phenomenon. It has a drag which can be measured.

Euler's equations are a model of it, parameter free, with geometry and forcing as data. They are answerable to the phenomenon: if the drag they give is wrong, nothing can be adjusted to repair it.

A proof about the equations is a statement about the model. It is answerable to the model alone.

The digital solution reaches all three levels. It is a solution of the equations, by the residual control; it is an object that can be inspected, and inspecting such objects is where the resolution of d'Alembert's paradox [9] and the account of flight [10] came from; and its mean values can be set beside a measurement, where they survive.

The symbolic proof reaches the first of the three and stops. That is not a criticism of it as mathematics, where it is an advance, and existence without an exhibited instance is ordinary and accepted practice — the disproof of the Mertens conjecture [18] established that a counterexample exists and exhibited none. It is a statement about what the result can be offered as. A theorem about a model becomes a statement about fluids only by way of an instance, and there is none here.

8 The shock, and the question it raises

The OpenAI proofs of singularity formation for Euler and for Navier–Stokes have sent a shock wave through the mathematical community, and the question being asked is plain enough. If a machine can produce proofs of theorems better than people can, what is left for those whose main occupation is proving theorems?

The question is legitimate and we do not propose to soften it. But two things about the present case bear on it, and both follow from the argument above.

What was automated was the derivation, not the judgement. The system worked on a problem already posed, already named, already agreed to be worth settling, and already being attacked by others along related lines [2]. It did not decide that finite-time blowup for Euler was the question to ask. It did not decide that Euler's equations were the right model of a fluid, which is a judgement about the world and not about symbols. And it did not connect its theorem to anything: the step from a statement about a model to a statement about fluids is not a proof step, and nothing in the output performs it. Sections 6 and 7 are an instance of that gap rather than a complaint about it.

And this has happened before, to the discipline in this paper. Machines outran hand calculation in the 1950s, and numerical analysis did not end. It changed what it was about. The object of study moved from finding a solution — which the machine would do — to bounding the error in what the machine produced, to deciding what was worth computing, and to establishing when a computed quantity is determined by the equations at all and when it is not. That is a harder subject than the one it replaced, and it is the subject that produced the error control of Section 4.

We would therefore expect the answer for mathematics to have the same shape, without pretending to know it. Deriving will be automated, and probably well. What will not be is choosing the question, judging whether a model deserves the attention, supplying the instance which lets a theorem touch something, and saying what a result means. On the evidence of this episode those are separate activities from derivation, because here they were left undone.

And there is relevant symbolic mathematics, which is not theorem-proving. If the proofs discussed here concern the model and not fluids, that is no reflection on symbolic

mathematics as such. It is a reflection on one use of it. The most consequential symbolic work in physics has never been the proving of theorems about given equations. It has been the *formulation of the equations*, from principles.

Equations (1)–(3) are themselves such a piece of work. Newton's second law is a statement about particles; writing it for a continuum, deciding that the force of the surrounding fluid is a pressure gradient, and adjoining incompressibility as a constraint with the pressure as its multiplier, is symbolic mathematics of the first order — and every step of it required deciding what a fluid *is*. The result is three lines carrying no parameters, from which everything in this paper follows. Schrödinger's equation is the same kind of achievement for atoms, and the reformulation of atomic physics as non-overlapping charge densities meeting at free interfaces, which one of us has pursued under the name RealQM [16], is an attempt at the same activity rather than at theorem-proving: the work is in what the equations should be, not in what follows from them.

This is where symbolic mathematics does its heaviest lifting, where physical judgement enters mathematics, and where, so far, no machine has done anything at all. A system which proves a theorem about Euler's equations has been handed the hard part.

9 Summary

The equations (1)–(3) are parameter free and have only geometry and forcing as data. The mathematical questions asked of them — existence, regularity, uniqueness — have gone unanswered in symbolic form because turbulent solutions have no structure for symbolic analysis to exploit, this being what turbulence is rather than a gap in technique. In digital form, by Numerical Euler, the same questions are answered, in altered shape: solutions exist by exhibition, are not regular, are not unique as fields, and are unique in the mean values anybody wants, with computed error bounds and agreement with measurement.

A symbolic proof of finite-time blowup settles a long-open question about the model. Its solution conserves energy exactly and so dissipates nothing, whereas a fluid dissipates and the dissipation is the drag; and it exhibits no individual case, so nothing can be inserted into the equations, inspected or measured. It is an exact statement about an object with no observables, set beside approximate statements about observables which agree with experiment. Each has what the other lacks, and only one of them is answerable to a fluid.

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